Housing First or Treatment First?

A stylized model

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Abstract

This section develops the smallest model in which the two main questions of the paper are still well posed. Who uses drugs, and why are users poor? And how does housing assistance change the decision to use? The model has three periods, one asset, and households that differ in permanent income and in nothing else. Because it is stylized, every magnitude reported in the paper comes from the quantitative model rather than from here. What this model delivers is a set of qualitative results: which feature of the environment reverses the income gradient in drug use, the sign and relative size of the two forces through which a housing voucher acts on drug use, and a dominance result showing that conditioning assistance on abstinence cannot raise welfare in an environment whose households are rational and whose drug use harms nobody else.

1 A stylized model

The quantitative model of the paper has six state variables and roughly forty parameters. That machinery is what the welfare and fiscal questions require, but it makes it hard to see which piece of the environment is doing the work. This section strips the environment down to the smallest one in which the two main questions of the paper are still well posed.

1. **Who uses drugs.** Drug use is concentrated among low income households. What features of the model produce it?
2. **How housing assistance changes decisions.** A voucher pulls drug use in two directions at once. The first is a moral hazard force, which raises use, because a household that acquires a substance use disorder is housed anyway. The second is a value of life force, which lowers use, because a housed future is worth more than a homeless one and using puts that future at risk. Which force is larger?

Because this model is stylized I do not use it to produce magnitudes, there is no validation section, and the calibration is deliberately rough: it exists to put the model in the right region, not to be defended. Throughout, each component of the environment is justified by removing it, recalibrating the model without it, and showing why it matters for one of the two results. Section 6 then asks the welfare question, and most of its answer is a statement about what an environment this small cannot contain: there is no government budget beyond the vouchers themselves, no externality from drug use, and no general equilibrium.

2 The environment

2.1 Households and preferences

There are three periods and a continuum of households. A household is characterised by its permanent income y, which is fixed over the household's life and lognormally distributed across households,

$$\log y \sim \mathcal{N}\left(-\tfrac{1}{2}\sigma^2,\ \sigma^2\right), \qquad \text{so that } \mathbb{E}[y] = 1. \tag{1}$$

Average income is therefore one, and every quantity below that is measured in units of income is a multiple of the population average. Households differ in y and in nothing else. There is no heterogeneity in preferences, so every household at a given income makes the same choice and drug use is a threshold in income rather than a probability. The sharpness is intentional: it lets the exercise in Section 4 report exactly where the threshold sits and how far it moves when a feature of the environment is switched off. It also keeps intact the paper's claim to generate the income gradient in drug use without assuming ex ante preference heterogeneity, since there is none here available to do the work.

Flow utility in a period depends on non durable consumption c, the flow of housing services h the household consumes that period, and drug consumption v,

$$u(c, h, v) \;=\; \log c \;+\; \gamma \log h \;+\; \theta\big[\log(b+v) - \log b\big] \;+\; \aleph. \tag{2}$$

The four preference parameters are the following.

- $\gamma > 0$ is the weight on housing services h. Housing enters as a single scalar index of quality and size, and the household chooses it in every period from the three point grid $\{h_0, h_S, h_L\}$ of equation (6): homelessness, a small unit, and a standard unit. Three options and no more, for the reason given in Section 2: two rented sizes are the minimum that lets a household economise on housing rather than lose it outright, and the third point is the outside option of having no home at all. The parameter γ governs how much utility a household loses when it moves down that grid, and therefore how much a housing voucher is worth.

- $\theta > 0$ is the weight on drug use. It and b govern the drug term jointly: the marginal utility of the vth unit is $\theta/(b+v)$, so θ mostly sets how much a user uses while b mostly sets whether anyone uses at all. Accordingly θ is set and b is calibrated.

- $b > 0$ is a Stone and Geary shifter on drugs. It is written this way for two reasons. First, the term is normalised to zero at $v = 0$, so a household that does not use drugs is neither rewarded nor punished for it and $\aleph$ below carries the entire flow value of being alive. Second, and more importantly, the marginal utility of the first unit of drugs is θ/b, which is finite. Under plain $\theta \log v$ that marginal utility is infinite at $v = 0$ and every household in the economy uses, whatever the consequences, so the model could not produce an interior participation rate at all. A smaller b means a stronger pull from that first unit.

- $\aleph > 0$ is the flow value of being alive: the utility a household receives in a period simply from surviving into it, over and above what it consumes. It is what gives mortality risk a cost that does not vanish when consumption is low, and it is why the mortality channel of Section 4 bites hardest at the bottom of the income distribution. Without it, a poor

household with low c and low h would have little to lose from dying and the model would produce the opposite gradient.

Households discount the future at rate β and save in a single risk free asset at gross return R, subject to a no borrowing constraint $a' \geq 0$.

2.2 Timing

Period 1. Every household is employed and has no substance use disorder. It chooses consumption, housing from the grid $\{h_0, h_S, h_L\}$, drug use $v \geq 0$, and saving. Housing is a choice here exactly as it is later, and the household pays the market rent on whatever it picks; the one thing that does not operate yet is the voucher, which arrives only from period 2. That costs the policy comparison nothing, because every period 1 household is clean, so Housing First and Treatment First would cover precisely the same households on precisely the same terms. Using drugs generates the risk of a substance use disorder in period 2, which rises with drug use v,

$$\pi(v) \;=\; 1 - e^{-\lambda v}, \qquad \lambda > 0. \tag{3}$$

The parameter λ is the addiction hazard per unit of use. It is calibrated to the share of users who acquire the disorder. The functional form matters for the household's problem: because π is increasing in v, the household internalises the fact that using more is riskier, so the intensive and extensive margins of drug use are not separate decisions.

Period 2. For households that used drugs in period 1, a substance use disorder $d \in \{0, 1\}$ may be realised, with probability $\pi(v)$; for households that did not use, it is not. The household is unemployed with probability u_0, or with the higher probability u_1 if the disorder is realised. An unemployed household receives income support, described in Section 2.3 below. The household then chooses consumption, housing, and savings. If the disorder is realised, the probability of recovering from it by the next period is ρ.

Period 3. The last period of life. Employment, income support and the voucher work exactly as in period 2, and the household again chooses consumption and housing from the same three point grid. There is no drug use decision: a household that does not already have the disorder has nothing to gain from acquiring one when there is no period 4 in which to bear the consequence, and a household that does have it consumes and houses itself as best it can. Assets are run down and nothing is bequeathed.

Mortality. A household survives from one period to the next with probability

$$\phi(d) \;=\; \phi_0 \,(1 - \kappa d), \tag{4}$$

so ϕ_0 is survival for a household without the disorder and κ is the proportional reduction in survival caused by it. This is the mortality channel, and it turns out to be the single most important feature of the model.

2.3 Employment and income support

A household is employed or unemployed in periods 2 and 3. The probability of being unemployed is u_0 without the disorder and $u_1 > u_0$ with it, so the gap $u_1 - u_0$ is the employment cost of the disorder. This is the entire resource cost of the disorder in the model. There is no wage penalty conditional on working: the quantitative model contains such a parameter and switches it off, because the evidence on earnings conditional on employment is contested while the robust findings are on the employment margin, and a stylized model should not rest on a shock that the model it introduces does not use.

Income in a period is

$$y_e = \begin{cases} y, & \text{if employed,} \\ \psi \cdot \min\{y,\ \bar{y}\}, & \text{if unemployed,} \end{cases} \tag{5}$$

where ψ is the replacement rate and $\bar{y}$ caps the earnings base on which the replacement rate is applied. A household at or below the cap therefore receives the fraction ψ of its own income, while a household above it receives $\psi\bar{y}$ regardless of how much it used to earn.

The level of income support is one of the most consequential numbers in the model, and the reason is worth stating. If income support were set above the rent on the smallest unit, every household could always afford a home, the disorder would never put anyone's roof at risk, a housing voucher would have nothing to insure, and the moral hazard force of Section 5 would be exactly zero. At the calibrated values it delivers half the income of the poorest households in the distribution, which is well below the rent on the smallest unit, so an unemployed household at the bottom cannot house itself out of current income alone and needs either accumulated savings or a voucher. That is the link between the disorder and homelessness, and without it the second result of this section does not exist.

The cap $\bar{y}$ does something different and more subtle, and it matters for the first result rather than the second. Without it, income support would be exactly proportional to income, and so would the entire resource cost of the disorder. A penalty that is proportional to income cannot move a gradient that income itself generates: it scales with precisely the thing it is meant to offset. The cap breaks the proportionality at the top, so that unemployment is a larger fall for a household above average income than for one below it, and it is what gives the unemployment channel of Section 4 any purchase on the income gradient at all. Real income support is capped in the same way and for the same reason.

2.4 Housing and the voucher

Rental housing is described by a single index of housing services h, and the rent per unit of housing services, p_r, is the numeraire, so a unit's size and its rent at market price are the same number. In periods 2 and 3 a household chooses among three options,

$$h \in \{h_0,\ h_S,\ h_L\}, \qquad h_0 < h_S < h_L, \tag{6}$$

a standard unit h_L, a small unit h_S, and homelessness h_0. Three points is the whole grid, and the count is not incidental: two rented sizes are the fewest that let a household economise on housing rather than lose it, and the third is the outside option of not renting at all. The first two are rented at $p_r h_S$ and $p_r h_L$. Homelessness is free: it costs nothing and delivers almost

nothing, h_0 being a small fraction of a rented unit. It is the outside option that a household falls back on when it cannot, or chooses not to, pay for a home.

Two rental sizes are needed rather than one, and the reason is quantitative rather than aesthetic. With a single size, a household whose cash on hand falls below the rent has no way to economise on housing at all and goes straight onto the street. Collapsing h_S into h_L raises homelessness from 0.36 per cent of the population to 3.29, and no value of h_0 brings it back: driving the street's housing services to essentially zero still leaves 0.81 per cent, twice the target, and refitting the whole model leaves 2.10 per cent, five times it. Making the street arbitrarily miserable does not help a household that cannot afford the rent at any level of misery. With a small unit available the household downsizes instead, which is what households do in the data and what the quantitative model's housing grid is for. Section 4 returns to what removing it does to the policy experiment.

A housing voucher covers part of the rent. It is means tested and rationed. A household is eligible in a given period if its income in that period is below the threshold y_V and if the policy in force covers its disorder status; conditional on eligibility a voucher arrives with probability π_V. Writing $z \in \{0, 1\}$ for whether a voucher is held,

$$\Pr\big(z = 1 \mid y_e, d;\ \text{policy}\big) \;=\; \pi_V \cdot \mathbf{1}\{y_e \le y_V\} \cdot \mathbf{1}\{\text{policy covers } d\}. \tag{7}$$

A household holding a voucher pays the fraction $\kappa_r < 1$ of the market rent, so the rent actually paid on option j is

$$r_j(z) \;=\; \begin{cases} 0, & j = h_0 \ \ \text{(homeless)}, \\ p_r h_j \big[\kappa_r z + (1 - z)\big], & j \in \{h_S,\ h_L\}. \end{cases} \tag{8}$$

The parameter π_V is the generosity of the programme: the share of eligible households actually covered. It is the policy dial traced out in Figure 2. Vouchers exist only in periods 2 and 3, when the disorder and unemployment have been realised and housing is a choice.

2.5 The two policies

The two policies differ in exactly one respect, which is the indicator $\mathbf{1}\{\text{policy covers } d\}$ in equation (7).

- **Housing First** covers every eligible household regardless of the household's disorder status, so the indicator is one for both $d = 0$ and $d = 1$.
- **Treatment First** covers only households that are clean, meaning those for which the disorder has not been realised ($d = 0$), so the indicator is one for $d = 0$ and zero for $d = 1$.

The only thing Housing First adds to Treatment First is cover in the disorder state. That is what makes the decomposition of Section 5 exact rather than approximate.

2.6 The household's problem

The problem is solved backward. Let a denote assets carried into a period, e the employment state with associated income y_e from equation (5), z the voucher state from equation (7), and d the disorder state.

In period 3 the household consumes everything and chooses housing only:

$$V_3(a,e,z) \;=\; \max_{j\in\{h_0,h_S,h_L\}} \Big\{ \log\big(Ra + y_e - r_j(z)\big) + \gamma \log h_j + \aleph \Big\}. \tag{9}$$

Taking expectations over employment and the voucher lottery, both of which depend on the disorder state, gives the value of entering period 3 with assets a and disorder status d,

$$W_3(a,d) \;=\; \mathbb{E}\big[\, V_3(a,e,z) \,\big|\, d \,\big]. \tag{10}$$

In period 2 the household chooses housing and saving, and faces both mortality risk and, if it has the disorder, the chance of recovery:

$$V_2(a,e,z,d) = \max_{j,\; a'\geq 0} \Big\{ \log\big(Ra + y_e - r_j(z) - a'\big) + \gamma \log h_j + \aleph + \beta\, \phi(d)\, \mathbb{E}_{d'}\big[W_3(a',d')\big] \Big\}, \tag{11}$$

where the disorder evolves as $d' = 0$ with probability ρ if $d = 1$ and $d' = d$ otherwise, so $\mathbb{E}_{d'}[W_3(a',d')] = (1-\rho d)\, W_3(a',d) + \rho d\, W_3(a',0)$, and $\phi(d)$ is survival from equation (4). The value of entering period 2 is again the expectation over employment and the voucher lottery, $W_2(a,d) = \mathbb{E}[V_2(a,e,z,d) \mid d]$.

In period 1 the household is employed and clean, and chooses housing, drug use and saving:

$$V_1(y) = \max_{j,\; v\geq 0,\; a'\geq 0} \Big\{ \log\big(y - r_j(0) - p_v v - a'\big) + \gamma \log h_j + \theta\big[\log(b+v) - \log b\big] + \aleph + \beta\, \phi_0\, \mathbb{E}_{d'}\big[W_2(a',d')\big] \Big\}, \tag{12}$$

where $r_j(0)$ is the full market rent of equation (8), since no voucher is held in period 1, p_v is the price of a unit of drugs, and the disorder arrives with the use dependent probability of equation (3), so that $\mathbb{E}_{d'}[W_2(a',d')] = \big(1-\pi(v)\big) W_2(a',0) + \pi(v)\, W_2(a',1)$.

A household uses drugs if the value in equation (12) at its optimal positive v exceeds the value at $v = 0$. Because households differ only in y, this comparison defines a set of income intervals on which use occurs, and the drug participation rate is the mass of the income distribution inside that set. Section 4 is about where those intervals sit.

3 Calibration

The calibration is rough by design and is presented in two parts. Table 1 lists the parameters set outside the model, either to standard values or directly to a measured quantity. Table 2 lists the three parameters calibrated inside the model, together with the moment each one is matched to and the fit.

One untargeted moment is worth reporting because it is the one the model gets least well. The share of the homeless who carry the disorder comes out at 0.56, which is the right order and the right direction but below what the data show. Nothing is calibrated to it and no result below turns on it; it is recorded here rather than buried because a reader who knows the homelessness literature will look for it.

One caveat belongs here rather than in a footnote. The three periods are stages of a life rather than calendar intervals, and the parameters in Table 1 take conventional values without imposing a mapping from a period to a number of years, so survival, recovery and the discount factor are not mutually consistent at any single horizon. Fixing that would require the life cycle

structure that the quantitative model has and this one deliberately does not. Nothing below turns on it: every comparison in Sections 4 and 5 varies one object and holds the rest of the parameter vector fixed, so the time unit is common to both sides of every number reported.

4 Who uses drugs

This section asks which feature of the environment determines who uses drugs across the income distribution. Start from the force that any such feature has to overcome. Drugs are a normal good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a disorder, that is the only force operating, and drug use concentrates at the top: the average user earns 3.41 times the population average, which is the first row of Table 3 and the left hand end of both curves in Figure 1.

Two features of the environment could reverse this gradient.

1. **The mortality channel.** The disorder kills you: survival falls by the factor κ in equation (4).
2. **The unemployment channel.** The disorder costs you a job with probability $u_1 - u_0$, and if income support cannot cover the rent you downsize and live with lower consumption, or lose your home altogether.

Each feature is scaled from nothing to its calibrated value, and at every point b is recalibrated so that the drug participation rate stays at 5.13 per cent. The *amount* of drug use is therefore identical all along both curves, and the only thing moving is *which* households use. Figure 1 plots the result, and Table 3 reports the same exercise at five points along each curve, together with the value of b that the recalibration selects.

Table 3 contains the main result of this section, and it is sharper than the statement that mortality matters more than unemployment. The mortality channel reverses the gradient at a quarter of its calibrated strength, taking the income ratio from 3.41 to 0.19, and it does not move again after that; the rest of the channel changes nothing about who uses. The unemployment channel does not reverse the gradient at all. Scaled to three quarters of its calibrated value it has barely moved the ratio, from 3.41 to 3.37, and even fully switched on it leaves users at 2.73 times average income, still well above the population mean. In this environment the unemployment channel *cannot* produce the concentration of drug use at the bottom of the income distribution, however strong it is made.

The b column of Table 3 shows the recalibration doing real work, and reading it is the best way to see why the exercise is set up this way. With no consequences a household needs almost no temptation to use, and $b = 0.226$ delivers a participation rate of five per cent. Once the mortality channel is fully on, the same five per cent requires $b = 0.0051$, more than forty times more temptation, because the consequences the household now faces have to be outweighed. Were b not recalibrated at every point, the curves in Figure 1 would be comparing economies with different amounts of drug use, and the comparison would say nothing about who uses.

Why mortality reverses the gradient and unemployment does not. The asymmetry is not obvious and it is worth stating carefully, because the same logic operates in the quantitative

Table 1: Externally set parameters

Parameter	Description	Source or rationale	Value
Preferences and demographics			
β	Discount factor	Standard	0.96
R	Gross interest rate	Standard	1.03
σ	Dispersion of log permanent income	US cross section	0.75
ϕ_0	Survival per period, no disorder	Life tables over the length of a model period	0.93
$\aleph$	Flow value of being alive	Set so that the value of a statistical life is of the right order	2.50
γ	Weight on housing services	Housing budget share	0.25
θ	Weight on drug use	Set; θ and b govern the drug term jointly and b is calibrated given θ	0.11
The disorder			
ρ	Recovery probability per period	Treatment episode data	0.40
κ	Proportional fall in survival with the disorder	Excess mortality of people with a substance use disorder	0.45
u_0	Unemployment probability, no disorder	BLS	0.049
u_1	Unemployment probability, with the disorder	NSDUH 2024, ages 26 and above, weighted	0.1625
Income support			
ψ	Replacement rate while unemployed	Unemployment insurance replacement rates	0.50
$\bar{y}$	Cap on the earnings base, multiple of average income	Benefit caps	1.00
Housing, drugs and the voucher			
p_r	Rent per unit of housing services	Numeraire	1.00
p_v	Price of a unit of drugs	Units of v are defined so that one unit costs one	1.00
h_L	Standard rental unit	Housing budget share, 21.2 per cent of average income	0.212
h_S	Small rental unit	Half the standard unit	0.106
κ_r	Tenant's share of the rent when covered	Voucher rules	0.30
y_V	Voucher means test, multiple of average income	Income limits	1.00
π_V	Voucher generosity in the baseline	Share of eligible households covered	0.27

Note: rent per unit of housing services is the numeraire, so a unit's size and its rent at market price are the same number and both are expressed as a share of average income. Average income is one by construction. Section 5 varies π_V from zero to one; the value in the table is the baseline used everywhere else.

Table 2: Internally calibrated parameters and their targets

Parameter	Description	Target moment	Data	Model	Value
b	Pull of the first unit of drugs	Drug participation rate	5.13%	5.13%	0.00394
h_0	Housing services when homeless	Homelessness rate	0.399%	0.356%	0.00013
λ	Addiction hazard per unit of use	Share of users who acquire the disorder	33.2%	32.8%	97.1

Note: the three parameters are calibrated jointly, but each is bracketed on the moment it moves most. The mapping is close to triangular: b shifts the drug participation rate with little effect on the other two, h_0 governs how unattractive the street is and therefore the homelessness rate, and λ converts a given amount of drug use into disorders. The calibration is rough on purpose and the fit is not defended anywhere in the paper; every magnitude reported in the paper comes from the quantitative model.

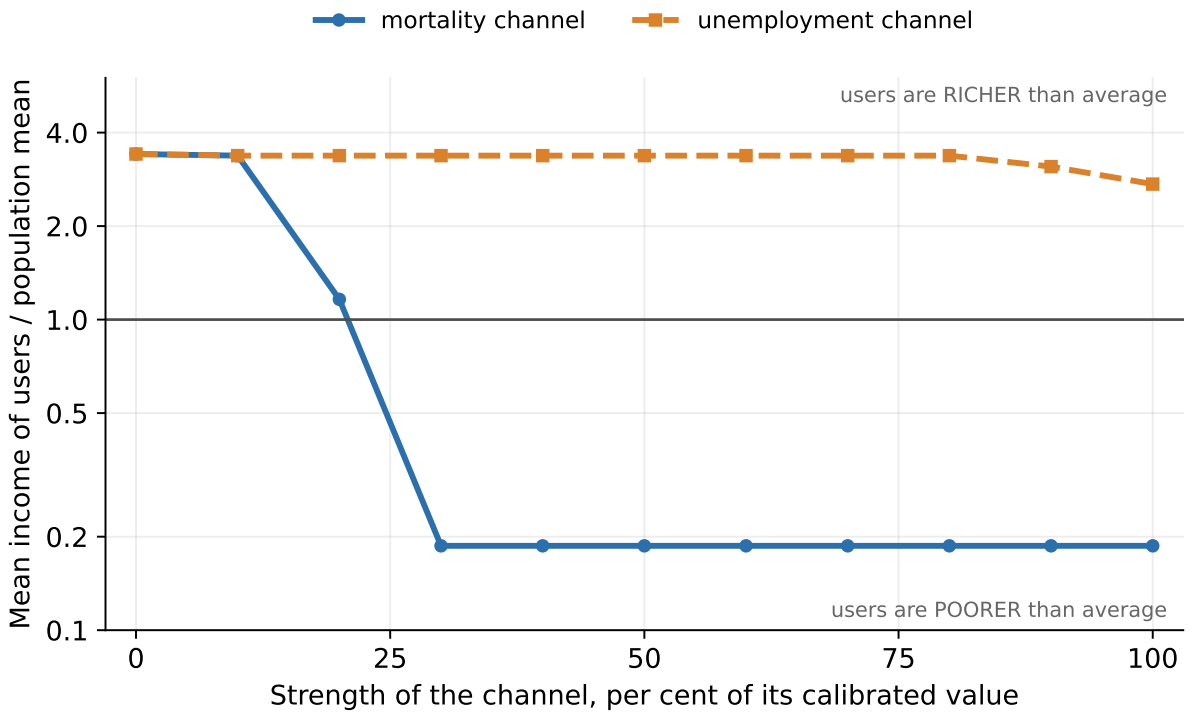


Figure 1: What it takes to reverse the income gradient in drug use

Note: the vertical axis is the average income of drug users divided by average income in the population, on a log scale; the horizontal axis scales each feature from zero to its calibrated value. Above the horizontal line at one users are richer than average, below it they are poorer. At every point b is recalibrated so that the drug participation rate is 5.13 per cent, so the two curves compare economies with the same amount of drug use and differ only in who does it. Both curves begin at 3.41, the case with no consequences at all.

Table 3: Who uses drugs, as each feature is scaled up

	Mortality channel		Unemployment channel	
Strength of the channel	b	$\bar{y}_u/\bar{y}$	b	$\bar{y}_u/\bar{y}$
0%	0.2260	3.41	0.2260	3.41
25%	0.0131	0.19	0.0593	3.37
50%	0.0086	0.19	0.0298	3.37
75%	0.0064	0.19	0.0165	3.37
100%	0.0051	0.19	0.0105	2.73

Note: $\bar{y}_u/\bar{y}$ is the average income of drug users divided by average income in the population. At each strength, b is recalibrated so that the drug participation rate is 5.13 per cent. The first row is the case with every consequence of the disorder removed and is common to both columns.

model. The unemployment channel is a penalty in resources, and a resource penalty scales with the very thing it is meant to offset: taking a fixed share of a large income still leaves a large income, so the penalty cannot outrun the fact that drugs are cheaper in utility terms for the rich. Indeed, if income support were exactly proportional to income the unemployment curve in Figure 1 would be essentially flat, running from 3.41 to 3.37 across its whole range, and the channel would be neutral on the income gradient by construction. What the cap $\bar{y}$ of equation (5) does is break that proportionality at the top, and the 2.73 in the last row of Table 3 is entirely the work of the cap. Even so, it is not enough.

The mortality channel is different in kind, because what it destroys is not resources but a continuation value, and the continuation value rises with permanent income. The household with more to lose loses more, and it loses proportionally more than it gains from drugs being relatively cheap. This is also why $\aleph$ in equation (2) is not decorative. It is the part of the continuation value that does not shrink as consumption falls, and it is what makes life worth living for a household with little to consume.

Cutting $\aleph$ to a fifth of its value does not distort the model, it inverts it. At $\aleph = 0.25$ the value of entering period 2 with no assets is negative for every household at or below average income, and for those households the value of entering *with* the disorder exceeds the value of entering without it: -0.68 against -0.70 at the mean. The disorder raises mortality, and a shorter life is preferable when life is worth less than nothing. Drug use stops being a risk taken for a benefit and becomes a way of ending a bad life. Participation rises to just over half the population, and raising b to 0.3, 1.0 or 3.0 leaves it above sixty per cent, so no value of b brings it back to five. No other parameter can do this, and it is why a flow value of being alive appears explicitly in equation (2) rather than being folded into the consumption term.

Why the mortality channel is retained. It carries both of the results in this section at once, which no other component does. Switching it off and recalibrating sends users back to 3.37 times average income, as Table 3 already shows. Less obviously, it also destroys the policy result of Section 5. Without mortality the value of life force is -0.23 percentage points instead of -6.45, which is to say it disappears: a voucher no longer deters anyone from using, because what using now threatens is only resources, and resources are what the voucher supplies either way. The moral hazard force survives at $+0.80$. An economy without the mortality channel would therefore conclude that Housing First unambiguously *raises* drug use, the opposite of the

conclusion reached here, and it would reach that conclusion with no opposing force in sight to warn the reader that the answer is quantitative. The name *value of life* is literal.

Why the unemployment channel is retained. The result just stated is a statement about the unemployment channel, so the channel has to be in the model for the statement to be made at all. It also does real work in Section 5. Removing it and recalibrating leaves the income gradient untouched at 0.19, confirming that it contributes nothing to who uses. What it does instead is carry most of the moral hazard: removing it cuts that force from +2.55 to +0.51 percentage points, four fifths of it, while the value of life force strengthens from −6.45 to −7.14. Unemployment is what makes the disorder threaten a household's housing, and housing is what the voucher insures, so without it a voucher has very little left to be a moral hazard about.

Why the small unit is retained. This is the one component whose removal breaks the model outright rather than merely moving a number, and it is why the ablation reports homelessness alongside the two forces. Judged on the forces alone the row looks harmless: the value of life force moves from −6.45 to −6.02 and the moral hazard from +2.55 to +2.72, and the verdict would be to remove it. But that row is not a recalibrated model. As Section 2 notes, a single rental size cannot deliver the homelessness rate at any value of h_0, and after refitting it sits at 2.10 per cent against a target of 0.399, five times too high. Without an intermediate option, a household whose cash on hand falls short of the rent goes straight onto the street, so the model has nowhere to put the households that in the data simply move somewhere smaller. Keeping a component because the model cannot hit its targets without it is a different kind of argument from keeping one because a result moves, and it is the stronger of the two.

Why saving is retained. Assets are the household's own insurance against the disorder, and they compete directly with the voucher. Shutting saving down and recalibrating leaves the value of life force almost unchanged, at −6.65 against −6.45 percentage points, but it drives the moral hazard from +2.55 to +4.74, which is nearly double. A household that cannot save has no way to protect its own housing, so a voucher is the only insurance there is and the behaviour it induces is correspondingly larger. Since the size of the moral hazard relative to the force opposing it is the object of Section 5, a component that changes it by this much cannot be dropped.

The generosity of income support. This is not a component that can be removed, since unemployment has to pay something, but its level can be varied. Halving the replacement rate, from $\psi = 0.5$ to 0.25, and recalibrating moves the value of life force from −6.45 to −5.40 percentage points and the moral hazard from +2.55 to +2.45. The deterrent force is the one that moves, by a full percentage point, because a household with less to fall back on has more at stake in keeping its home and therefore values the voucher more; the moral hazard barely notices. The ranking is untouched, so the level of income support is not a knife edge in the middle of its range. Its *upper* end is, for the reason given in Section 2.3: set income support above the rent on the small unit and the moral hazard force is exactly zero.

Recovery, and a component retained for a reason that is not quantitative. Setting the recovery probability ρ to zero and recalibrating moves the value of life force from −6.45 to −6.13 and the moral hazard from +2.55 to +3.45, neither of which changes anything qualitative,

so by the standard applied to everything else this would be grounds for removing it. It is retained anyway, because without recovery a household that ever acquires the disorder has it permanently, and Treatment First becomes a policy of permanent exclusion rather than a policy that conditions assistance on getting clean. Recovery is what gives the name of the policy any content. This is a modelling reason and not a quantitative one, and it is stated as such.

Why there are three periods and not two. A third period looks like an obvious economy until one asks what it does. Mortality is survival from period 2 to period 3, so deleting period 3 does not shorten the model, it deletes the mortality channel. A genuine two period version therefore has to move the timing: the disorder is realised at the end of period 1, survival into period 2 is $\phi(d)$, and period 2 is terminal. Recovery has nowhere left to act, but everything the two results rest on is still there.

Built and calibrated, that version fits marginally better than this one and delivers Result 1 more sharply: mortality still reverses the gradient, from 3.37 to 0.19 by half strength, and the unemployment channel is now exactly flat at 3.37 at every strength, never moving at all. Result 2 does not survive. The value of life force is -5.73 percentage points, close to the -6.45 here, but the moral hazard collapses from $+2.55$ to $+0.24$, a factor of ten. Two first order forces that nearly cancel becomes one force and a rounding error, and the central claim of Section 5 is gone.

The reason is worth stating because it is what the third period is for. The moral hazard is the value of being insured in the disorder state, and with a single terminal period after the disorder there is very little to insure: the household holds the disorder for one period rather than two, and because mortality now strikes before that period rather than after it, a household that acquires the disorder frequently does not survive to collect the voucher at all. Three periods are the fewest that give the disorder a future long enough to be worth insuring against.

Components that were tried and removed. Two further consequences of homelessness were built into an earlier version of this model and then removed, because after the model was fully recalibrated without them neither result moved: that living on the street raises mortality on top of the disorder, and that recovery from the disorder is harder on the street. Both belong in the quantitative model, where mortality and welfare are the objects of interest and the counterfactuals move the homeless population a long way. Here they cannot matter, for an arithmetic reason: homelessness is four tenths of one per cent of the population, so a component acting only on the homeless cannot move a population aggregate. A supply curve for drugs was tried and removed on the same test, changing neither result to two decimal places. A direct wage penalty for the disorder was removed for a different reason, given in Section 2.3: the quantitative model does not use one.

A caution about this kind of test. Removing a component and recalibrating is necessary but it is not sufficient, and the cap $\bar{y}$ is the example. Judged at the baseline it is worthless: participation, the income gradient and both policy forces are identical to four decimals with and without it, because at the baseline the mortality channel is on, drug users are poor, and the cap never binds on anyone who is deciding anything. It survives only because Figure 1 scales the unemployment channel up on its own, which is the one place in this section where the households near the margin are rich. A component has to be tested where the results actually put it.

5 How housing assistance affects drug use

A housing voucher acts on the decision to use drugs through two forces that work in opposite directions.

1. **The value of life force, which lowers drug use.** A voucher raises the value of remaining clean, because a housed future is worth more than a homeless one and using drugs puts that future at risk. In terms of Section 2.6, it raises $W_2(a', 0)$ in equation (12), the continuation value the household keeps by not acquiring the disorder.

2. **The moral hazard force, which raises drug use.** A voucher also cushions the consequence of acquiring the disorder, because a household that acquires it is housed anyway. It raises $W_2(a', 1)$, the continuation value the household falls back on if the risk it takes goes badly.

The difficulty in measuring either force is that any real programme moves both at once. The two policies of Section 2 separate the two channels exactly. Under Treatment First the voucher is withdrawn on acquiring the disorder, so $W_2(a', 1)$ is unchanged and only the value of life force operates. Under Housing First it arrives regardless, so both channels operate. Since cover in the disorder state is the only difference between the two policies, the gap between them is the moral hazard force, and the whole of it.

Figure 2 traces both policies as the generosity of the programme, π_V, rises from zero to one, and Table 4 reports the two end points and the decomposition.

Table 4: The two forces at universal cover

	Participation rate	Difference
Panel A. Drug use under each policy		
No voucher programme ($\pi_V = 0$)	6.45%	
Universal Treatment First ($\pi_V = 1$)	0.00%	
Universal Housing First ($\pi_V = 1$)	2.55%	
Panel B. The two forces		
Value of life (Treatment First − no programme)		−6.45 pp
Moral hazard (Housing First − Treatment First)		+2.55 pp
Net effect of Housing First (Housing First − no programme)		−3.90 pp

Note: all differences are in percentage points of the population drug participation rate, measured against the same baseline of no voucher programme. Universal cover means $\pi_V = 1$, so every eligible household receives a voucher; eligibility still requires income below y_V, and under Treatment First it also requires being clean.

Table 4 reads as follows. With no programme at all, 6.45 per cent of households use drugs. Moving to universal Treatment First, participation falls by 6.45 percentage points, to nothing: guaranteeing housing to households that stay clean makes staying clean worth a great deal more. Moving instead to universal Housing First leaves participation at 2.55 per cent, which is 2.55 percentage points higher than under Treatment First at the same generosity, and that difference is the moral hazard force. The net effect of universal Housing First against no programme is

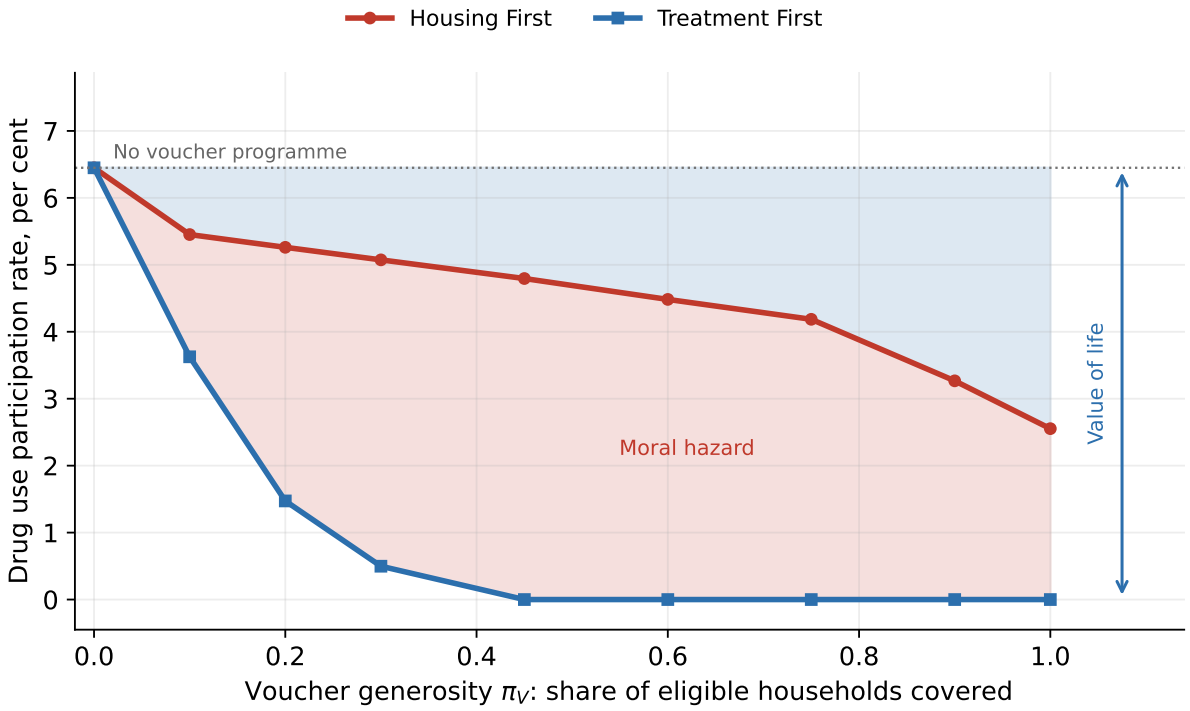


Figure 2: Drug use under the two policies, as the voucher becomes more generous

Note: the horizontal axis is π_V, the share of eligible households covered by a voucher, from no programme at the left to universal cover at the right. The vertical axis is the drug participation rate in the population, and the dotted line marks that rate with no programme at all. The red band between the two policies is the moral hazard force, the extra drug use that Housing First produces relative to Treatment First at the same generosity. The blue band above it is what Housing First delivers net of that moral hazard. The two bands together are the value of life force, marked by the vertical arrow at the right hand edge: the fall in drug use that Treatment First produces relative to no programme.

−3.90 percentage points: it still lowers drug use, because the value of life force is larger than the moral hazard force.

What the same policy dial does to homelessness and to the disorder. Drug use is not what a housing programme is for, and a reader shown only Figure 2 could reasonably conclude that Treatment First wins on every count. Figure 3 plots the two outcomes the programme is actually aimed at, over the same range of π_V.

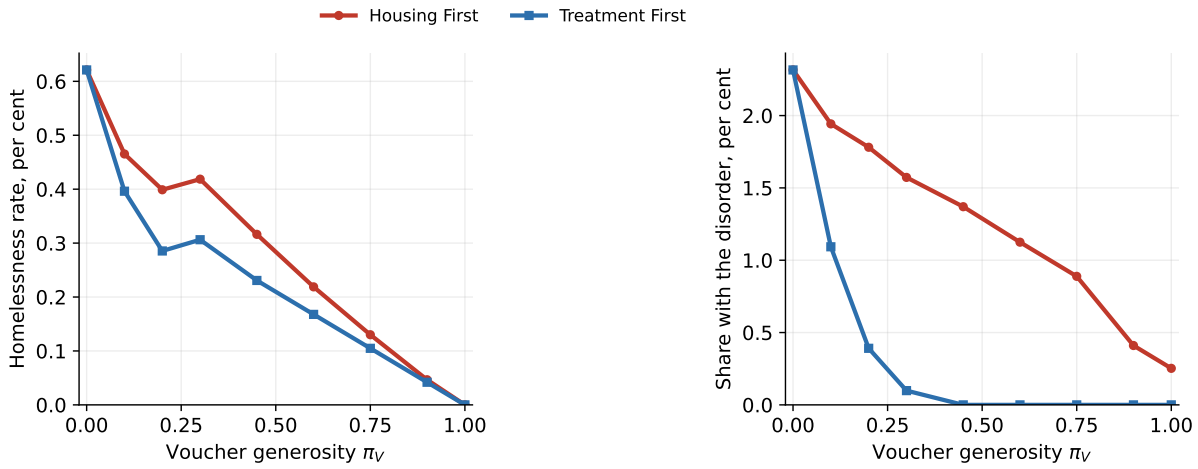


Figure 3: Homelessness and the disorder under the two policies

Note: both panels use the same horizontal axis as Figure 2, the share π_V of eligible households covered. The left panel is the homelessness rate in the population, the right panel the share of the population carrying a substance use disorder. Because households here are identical at a given income, decisions are sharp and the use region moves in steps as π_V rises, which is what produces the small kinks in the left panel.

Two things stand out. On homelessness the two policies end in the same place: both drive it from 0.62 per cent to zero at universal cover, and the gap between them along the way never exceeds about a tenth of a percentage point. Housing First does not fail at housing people. It houses them exactly as effectively as Treatment First does, which is what one should expect of a programme whose only difference is that it also covers households in a state that makes them poorer.

The difference is entirely on the other margin. At universal cover Housing First leaves 0.25 per cent of the population with the disorder and Treatment First leaves none. So the moral hazard of Section 5 does not show up as families left on the street; it shows up as disorder cases among people who are housed. That distinction matters for the welfare question of Section 6, because it means the two policies differ in what happens to a small group of households rather than in whether the programme achieves its housing objective.

Three things follow, and they are the reason the paper needs a quantitative model.

Both forces are first order, and they work against each other. The value of life force is about two and a half times the moral hazard force here, but both are large relative to the

participation rate they act on, and their near cancellation is the central feature of the problem. Theory does not sign the effect of housing assistance on drug use. Anyone asserting that housing the homeless must increase drug use, and anyone asserting that it must reduce it, is making a quantitative claim that needs to be tested while sounding as though they are making a theoretical one.

Unconditional assistance still lowers drug use. This runs against the intuition that unconditional help must raise drug use through moral hazard. The moral hazard here is real and it is large, at +2.55 percentage points. It simply loses to the force running the other way. Which of the two wins, however, is a quantitative question, and the ranking obtained in this stylized environment is not something the paper carries forward as a result; it is a hypothesis for the quantitative model to test.

Conditionality retains the winning force and removes the losing one. That is the case for Treatment First *as a way of reducing drug use*, and it is worth being careful about what it is not: Section 6 shows that in this environment the same policy makes every household weakly worse off, so a lower participation rate is not by itself an argument for the policy that produces it. Figure 2 also shows the condition under which even the drug use case is weak. Both forces scale with π_V: at the left of the figure the two policies are almost indistinguishable, because a programme that reaches few households has little moral hazard for conditionality to remove. Imposing conditionality while holding the number of vouchers fixed at a low level therefore buys very little, while the cost of excluding people from a scarce benefit is immediate. Conditionality and generosity are complements, and an experiment that changes one while holding the other fixed will understate what conditionality can do.

6 Welfare

Nothing so far has said whether Housing First or Treatment First is the better policy, only what each does to drug use. The natural next question is whether the moral hazard force of Section 5 is a reason to prefer conditionality. In this model it is not, and the reason is worth setting out because it is an argument rather than a number.

Housing First dominates Treatment First at equal generosity. Compare the two policies at the same π_V. By equation (7), they differ only in the indicator $\mathbf{1}\{\text{policy covers } d\}$: both pay the voucher when $d = 0$, and only Housing First pays it when $d = 1$. Housing First therefore transfers resources to the household in a superset of the states in which Treatment First does, so the household's budget set is weakly larger in every state and strictly larger in one. A household that optimises cannot be made worse off by that, and is strictly better off whenever it faces any chance of the disorder:

$$V_1(y;\ \text{HF},\ \pi_V)\ \geq\ V_1(y;\ \text{TF},\ \pi_V) \qquad \text{for every } y, \tag{13}$$

with equality exactly for households that never use. This is a dominance argument, not a calibrated result. It does not depend on any parameter and no choice of parameters can overturn it. Table 5 confirms it: 3.6 per cent of households, which is to say the ones who use, are strictly worse off under Treatment First, and no household anywhere in the distribution is better off.

The moral hazard is not a distortion. This is the part worth pausing on, because the language invites the opposite reading. Households here are forward looking, face the true addiction hazard of equation (3), and make no mistakes. The extra drug use that Housing First induces is therefore not a wedge to be closed. It is a household optimally taking a risk that it is now better able to bear, in exactly the sense in which an insured household drives in worse weather. The behaviour identified in Section 5 is real and it is large, but its size is a statement about drug use and not, on its own, a statement about welfare.

Table 5: Welfare and cost at universal cover

	Housing First	Treatment First
Panel A. Outcomes at universal cover		
Drug participation rate	2.55%	0.00%
Share with the disorder	0.253%	0.000%
Homelessness rate	0.000%	0.000%
Voucher outlay per household	0.1791	0.1792
Panel B. Treatment First against Housing First, no externality		
Households strictly worse off		3.6%
Households strictly better off		0.0%
Consumption equivalent loss, those affected		0.30%
Consumption equivalent loss, whole population		0.011%
Panel C. How large an externality would reverse this		
Disorder cases avoided by conditionality		0.253 pp
Critical social cost per case		0.12

Note: universal cover means $\pi_V = 1$ under both policies. Voucher outlay is the expected present value of the government's spending on a household from period 2 onwards, discounted at R and weighted by survival, expressed as a multiple of average income. The consumption equivalent is the proportional change in consumption, in every period and state, that would be worth as much to the household as the difference in lifetime value; it is reported both for the households that are affected at all and for the whole population.

Matching the two policies on cost does not change the ranking either. Equation (13) holds generosity fixed, which is the wrong comparison if vouchers must be paid for. The natural conjecture is that Housing First buys the same coverage at a higher price, since it pays for households in the disorder state and, by insuring that state, puts more households into it; a fixed budget would then buy more coverage under Treatment First, and that would be a reason to prefer it. The conjecture is false here. Treatment First is very slightly *more* expensive at every generosity, 0.1792 against 0.1791 at universal cover, so matching the two policies on cost rather than on π_V leaves Treatment First welfare reducing and still leaves no household better off.

Why the additional coverage is nearly free. A household in the disorder state draws only about 79 per cent of the voucher outlay a clean household draws, and this holds at every income level. It is not that such households house themselves worse: Figure 3 shows homelessness at zero under both policies at universal cover. It is mortality. Survival to period 3 falls from $\phi_0 = 0.93$ to $\phi_0(1-\kappa) = 0.51$, so the programme pays for these households over fewer periods, and the saving very nearly offsets the cost of covering them. The mortality channel, which Section 4

showed to be what puts drug use at the bottom of the income distribution, turns out also to be what neutralises the fiscal case against insuring it.

How large an externality would be needed. What is left is the possibility that drug use harms someone other than the user. Suppose each household carrying the disorder imposes a social cost χ on the rest of society, in units of one household's lifetime income, and that this cost is invisible to the household itself, which is what makes it an externality rather than something already priced into equation (12). Conditionality is then worth having when the cases it prevents are worth more than the insurance it withdraws.

Panel C of Table 5 reports the critical value. Universal conditionality prevents 0.253 percentage points of disorder cases at a private welfare cost of 0.011 per cent of consumption, so the two are equal at

$$\chi^{\star} \approx 0.12 \tag{14}$$

of a household's lifetime income per case avoided. This is the most useful number the section produces, because it is a magnitude a reader can weigh rather than a direction. It is well within the range that crime, emergency care and lost output to others could plausibly justify, which is precisely why the question cannot be settled here: the model has no crime, no health system, and no one but the user. What it can say is that the private welfare cost of conditionality is small enough that a moderate externality overturns the ranking, so everything depends on a magnitude that has to be measured elsewhere.

What this leaves for the quantitative model. Two of the four candidate justifications for conditionality can now be dismissed inside this model rather than merely named. Matching the two policies on cost rather than on generosity does not help, because they cost the same; and a government budget constraint does not help either, for the same reason, which is why there is none here. What survives is the following, and the ordering is by how much of the answer each one is likely to carry.

1. **Externalities.** This is the one the model can size, and equation (14) sizes it: a social cost of roughly 0.12 of a household's lifetime income per case of the disorder is enough to reverse the ranking. Drug use here touches nobody but the user, and once it raises crime, public disorder, or health costs borne by others, the private optimum stops being the social one.

2. **Internalities.** Households here face the true hazard and discount the future consistently. If drug use involves present bias, or a household that underestimates λ, then the extra use is a mistake by the household's own lights and conditionality can be paternalistically justified. This model cannot speak to that, because it assumes the mistake away.

3. **Public costs outside housing.** The budget here is vouchers and nothing else, which is why the financing exercise comes out flat: the only fiscal consequence of a disorder case in this model is that the household draws its voucher for fewer years. Healthcare, treatment and incarceration all rise with the disorder and none of them appear. This, rather than the housing budget, is where the fiscal argument really lives, and it is a form of the externality above rather than a separate mechanism.

4. **General equilibrium.** With an endogenous drug price and a government budget that has to balance, both policies change prices and taxes for households that never touch either programme.

All four are present in the quantitative model, which is the sharpest statement this section can make about why that model is needed: not to refine the magnitudes of Section 5, but because the welfare question cannot be posed at all in an environment this small.

7 Summary

Three results carry forward, and they carry forward as different kinds of object. The first is a statement about mechanisms: drug use concentrates at the bottom of the income distribution only when the disorder carries a mortality cost, because a resource penalty proportional to income scales with the thing it is meant to offset while a mortality penalty destroys a continuation value that rises with income. The second is a statement about the policy problem: a housing voucher acts on drug use through two first order forces that work against each other, and the two policies separate them exactly. That both forces are large is what this section establishes. Which of them wins is left open, and settling it is the job of the quantitative model that follows.

The third is a statement about what the model cannot settle, and it is the one that says most clearly why the quantitative model is needed. Because households here are forward looking and face the true addiction hazard, and because drug use harms nobody but the user, Housing First dominates Treatment First for every household at any given generosity, and matching them on cost rather than on generosity does not disturb that, because the two policies cost the same. What is left is an externality, and the model can say how big it would have to be: a social cost of roughly an eighth of a household's lifetime income per case of the disorder. The stylized model cannot rank the two policies on welfare, but it can say precisely which absent ingredient any such ranking must come from, and how large that ingredient has to be before it decides the question.